

COMP 3311

Database Management Systems

Lab 1

Oracle Data Modeler

Lab Topics

- How to construct an E-R diagram using [Oracle Data Modeler](#).

Oracle Data Modeler

- ❑ **Oracle Data Modeler** is a database design tool that allows you to:
 - create, browse and edit E-R diagrams;
 - reduce an E-R diagram to a relational schema (we won't be using this functionality in this course).

- ❑ **Oracle Data Modeler** is included with **Oracle SQL Developer**, which can be downloaded from <https://www.oracle.com/database/sqldeveloper/technologies/download/>
 - Windows, Mac, Linux available (requires JDK 11).
 - For Windows you should download the 64-bit version that includes JDK 11.

Oracle SQL Developer (1)

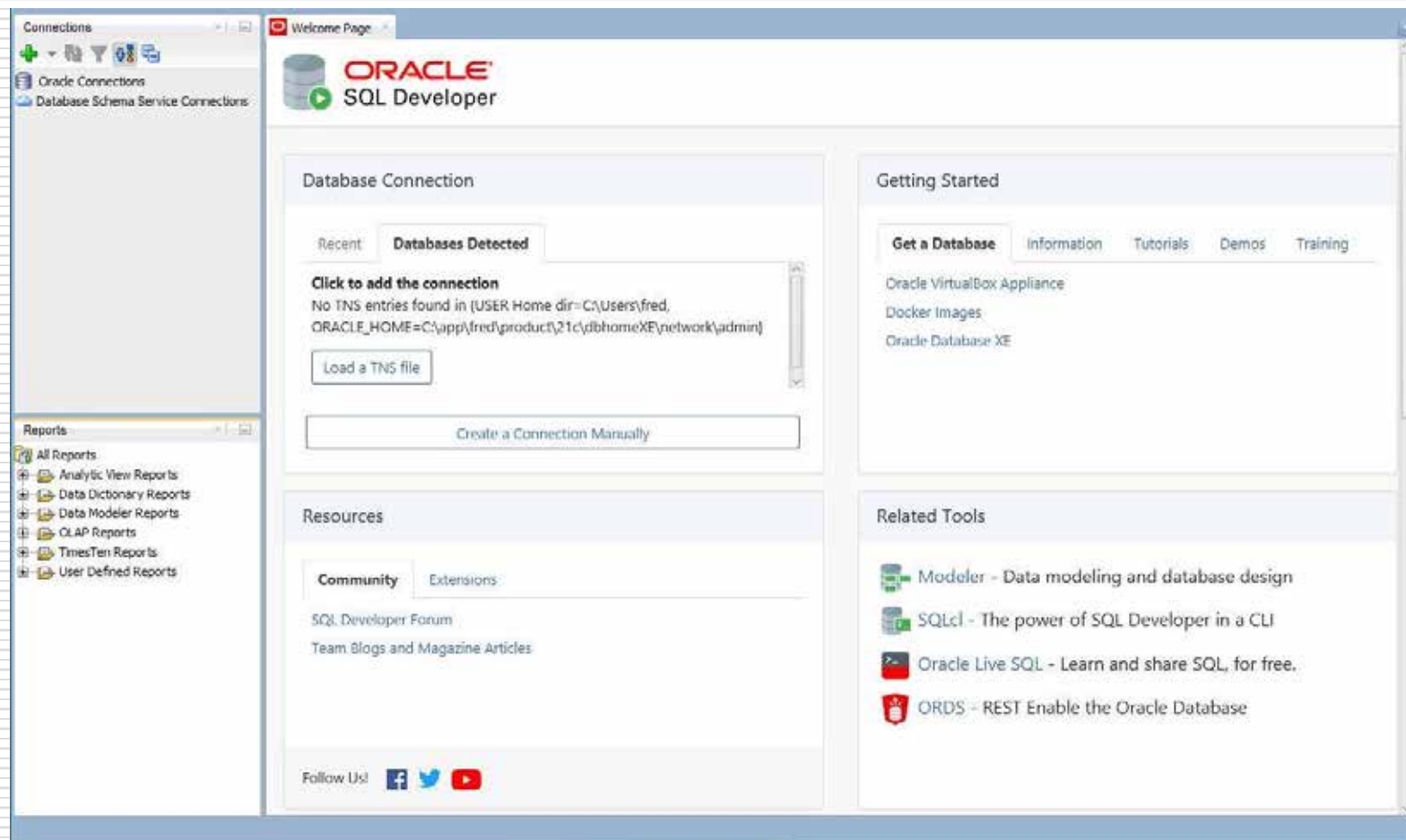
- Run the program “[sqldeveloper](#)”.

Double click the app.

SQL Developer opens in the [Welcome Page](#) as shown on the next slide.

Oracle SQL Developer (2)

- ❑ Close the [Connections](#), [Reports](#) and the [Welcome Page](#) tabs (the [Connections](#) tab will be discussed in the next lab).



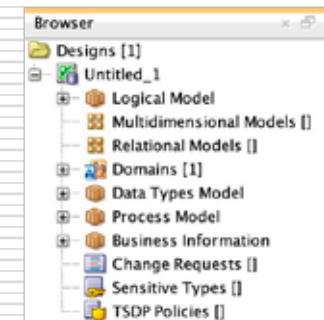
Create A Database Design

- To create a database design:

- select View→Data Modeler→Browser in the Oracle SQL Developer menu to open the Browser tab (see figure on the right);

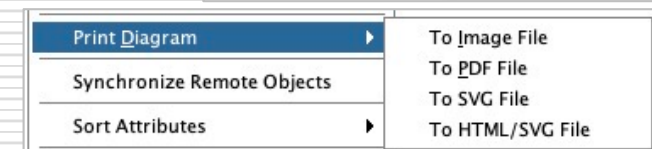
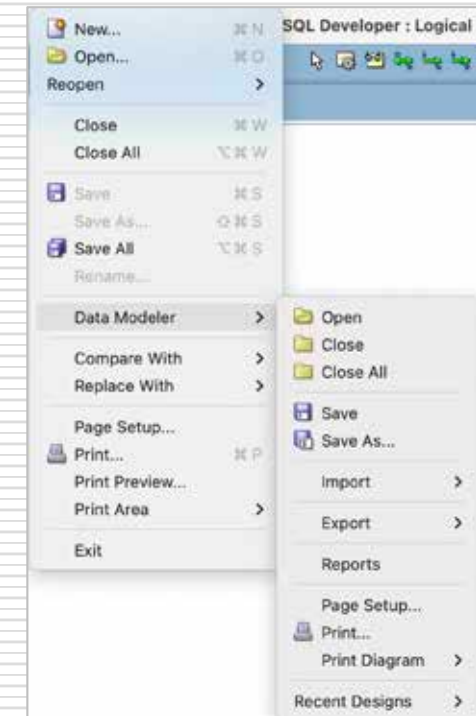


- The Browser tab contains a default template called Untitled_1 for creating a database design using the E-R data model (see figure on the right).
 - expand the Untitled_1 entry (if necessary).
 - right-click on the Logical Model node.
 - select Show from the popup menu to open the design surface.
- This template will be renamed when you save your design as explained on the next slide.



Save, Open, Print A Database Design

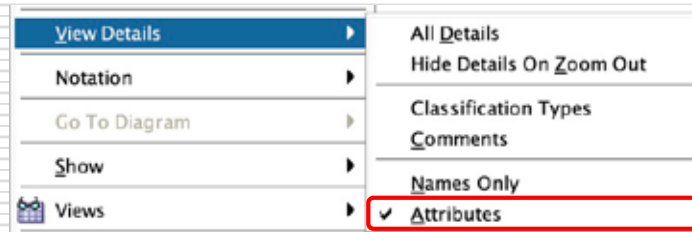
- ❑ To save a design (see figure on the right):
 - select File→Data Modeler→Save or File→Data Modeler→Save As... in the Oracle SQL Developer menu;
- ❑ To open a saved design (see figure):
 - select File→Data Modeler→Open in the Oracle SQL Developer menu;
 - navigate to the folder containing the design and select its .dmd file.
- ❑ To print a design:
 - right-click in the design surface;
 - select Print Diagram from the popup menu;
 - select one of the options shown in the figure above.



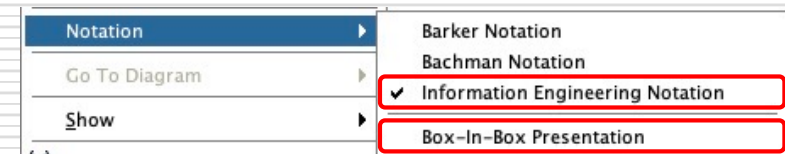
Oracle Data Modeler Design Settings

- ❑ Right-click in the design surface of the **Logical** tab (i.e., the design surface) and make the following selections from the popup menu for the indicated entries.

- **View Details:** check only
Attributes

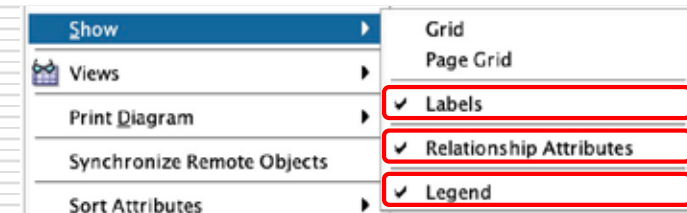


- **Notation:** check Information Engineering Notation;
uncheck Box-in-Box Presentation



Note
If options other than **Attributes** is selected in **View Details**, **Oracle Data Modeler** will show many additional details about entities that are not relevant for the purpose of constructing only an E-R diagram.

- **Show:** check Labels, Relationship Attributes, Legend¹



1. Right-click the Legend in the design surface, select Properties and uncheck the entries that you do not want to display.

Oracle Data Modeler Toolbar



- ❑ The **Oracle Data Modeler** toolbar contains buttons for the following operations (among others):
 -  **Select** allows selection of diagram elements.
 -  **New Entity** creates an entity type.
 -  **New N:M Relation** creates an N:M relationship type.
 -  **New 1:N Relation** creates a 1:N relationship type.
 -  **New 1:N Relation Identifying** creates a 1:N identifying relationship type for a weak entity.
 -  **New 1:1 Relation** creates a 1:1 relationship type.
 -  **New Arc** creates an XOR (exclusive or) constraint.
 -  **New Note** creates a note.

Create An Entity Type ⁽¹⁾

- Select the **New Entity** button  and click anywhere in the design surface.

Note: Until another toolbar button is selected, the **New Entity** button is active, and several entities can be created consecutively by simply clicking in the design surface.

- In the **Entity Properties** dialog, shown on the next slide:
 - enter a name for the entity type in the **Name** field.
 - select the **OK** button.
- An entity appears on the design surface like those shown on the slide [Create An Entity Type ^{\(3\)}](#).

Create An Entity Type (2)

Entity Properties - Employee

General

Name: Employee

Short Name:

Synonyms:

Synonym to display:

Preferred Abbreviation:

Long Name: Employee

Based on Structured Type:

Super Type: Select

Source:

Allow Type Substitution: ☒

Create Surrogate Key: ☐

Deprecated: ☐

OK Apply Naming Rules Cancel Help

Create An Entity Type (3)

Reposition an entity by selecting it and dragging it to the desired position.

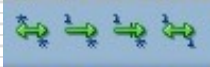
Resize an entity by selecting it and dragging one of its handles.

To better help align entities you can:

- select **Layout**→**Snap to Grid** in the design surface;
- select **Show**→**Grid** to show a grid.



Create A Relationship Type ⁽¹⁾

- ❑ Select a **Relation** button  and click inside one entity (the source) and then click inside the other entity (the target).

Note 1: Until another toolbar button is selected, the selected **Relation** button is active, and several relationships can be created consecutively by selecting the source and target entities.

Note 2: To create an identifying relationship either select the **New 1:N Relation Identifying** button in the toolbar or check the Identifying checkbox in the **Relation Properties** dialog.

- ❑ In the **Relation Properties** dialog shown on the next slide:
 - enter a name for the relationship type in the **Name** field.
 - edit the cardinality and participation constraints, if necessary.
 - check the **Identifying** checkbox if the relationship is identifying (or select the **New 1:N Identifying relationship** button initially).
 - select the **OK** button.
- ❑ A relationship appears on the design surface like those shown on the slide [Create A Relationship Type \(4\)](#).

Create A Relationship Type (2)

Relationship names do not display in **Oracle Data Modeler**.

- ❑ Instead, **Oracle Data Modeler** allows role names to be displayed, which are labelled **Name on Source** and **Name on Target** in the **Relation Properties** dialog (see the slide [Create A Relationship Type \(3\)](#)).
 - ❑ A workaround to display relationship names is to enter a relationship name into only one of the **Name on Source** or **Name on Target** fields in the **Relation Properties** dialog and enter a space into the other field as shown on the slide [Create A Relationship Type \(3\)](#).
- Note:** If only one role name has a value, then the name will not be displayed.
- ❑ The relationship name then displays at one end of the relationship, as shown on the previous slide, and cannot be moved from its location.

Create A Relationship Type (3)

Relation Properties - WorksOn

General

Name: WorksOn

Use surrogate keys: ☐

Source Cardinality

Source: Employee

Source key:

Name on Source: WorksOn

Source Entity Synonym: Employee

Source to Target Cardinality: 1 to *

Source Optional: ☒

Transferable: ☒

Target Cardinality

Target: InternalProject

Target key:

Name on Target:

Target Entity Synonym: InternalProject

Target to Source Cardinality: * to 1

Target Optional: ☒

Transferable: ☒

Dominant Role: None

Identifying: ☐

In Arc: ☐

Delete Rule: NO ACTION

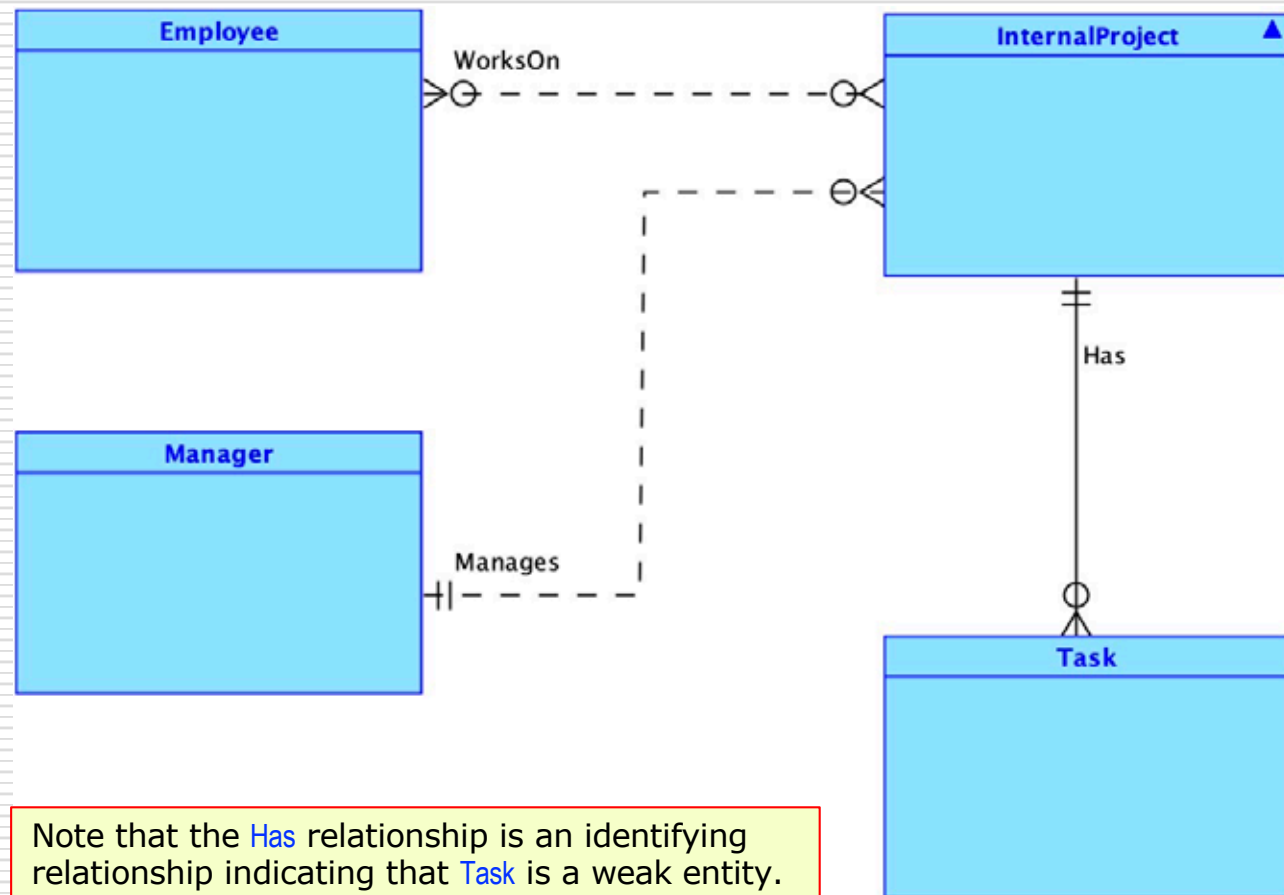
OK Apply Cancel Help

enter only a space in this field

Create A Relationship Type (4)

Reposition a relationship by selecting it and dragging its endpoints to the desired positions.

Add elbows to a relationship by right-clicking on it, selecting **Add Elbow** from the popup menu and dragging the elbow to the desired position.



Note: To create an identifying relationship, such as the one between `InternalProject` and `Task`, either select the **New 1:N Relation Identifying** button in the toolbar when creating the relationship or check the **Identifying** checkbox in the **Relation Properties** dialog for the relationship.

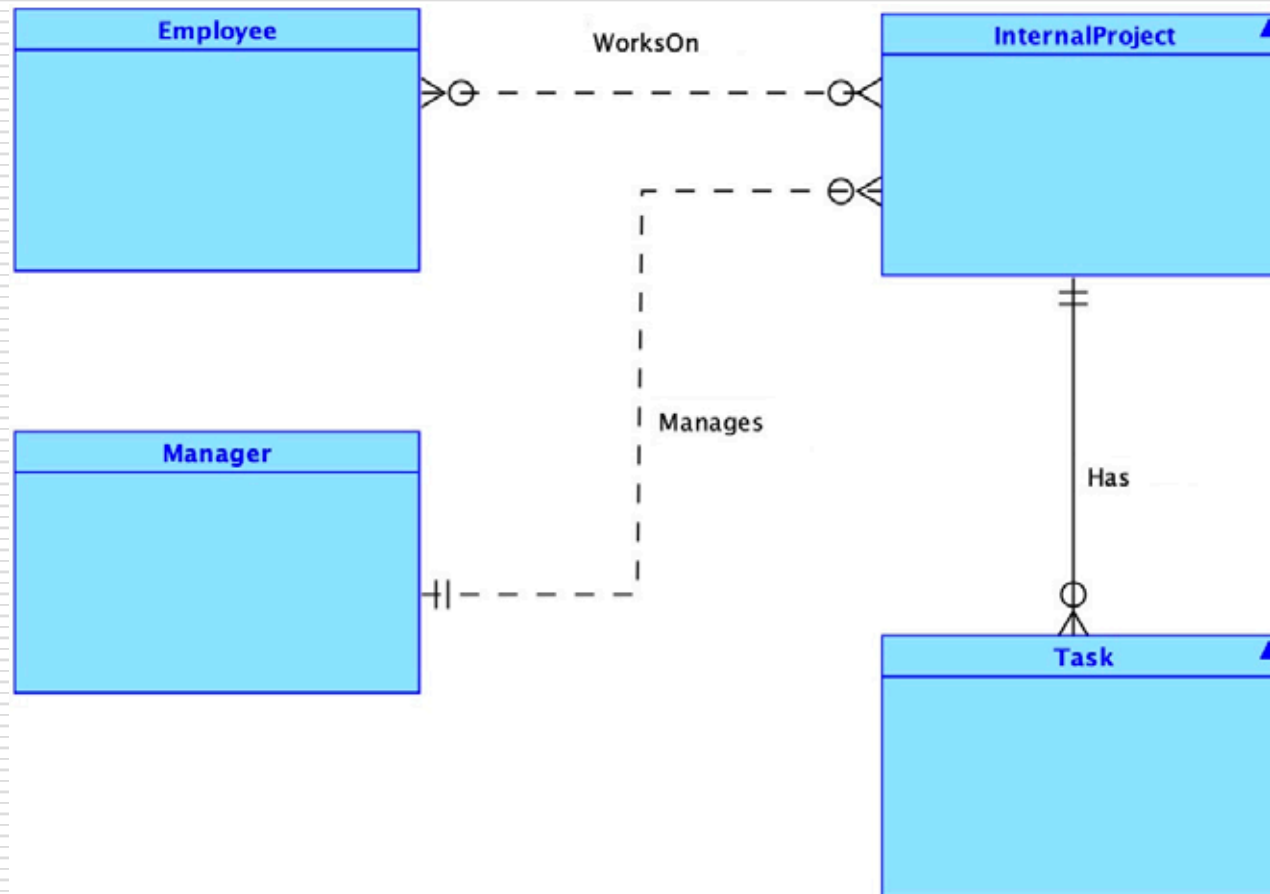
Create A Relationship Type (5)

- Alternatively, relationship names can be added manually via a **New Note** element as shown on the slide [Create A Relationship Type \(6\)](#).

Note: It is advisable to add relationship names using notes only after completing the diagram to avoid having to move the notes as relationships are moved.

- A note can be moved to the desired location by selecting and dragging it.
- To remove the background colour and border of a note, right-click it, select **Format** from the popup menu and
 - unselect **Use Default Format**.
 - set the **Background Color** and **Border Color** to white.

Create A Relationship Type ⁽⁶⁾



Add Entity Attributes (1)

- ❑ Open the **Entity Properties** dialog for an entity (shown on the next slide) either by double clicking it or right-clicking it and selecting **Properties** from the popup menu.

- ❑ In the **Entity Properties** dialog:
 - select the **Attributes** tab in the left column.
 - select **+** to add a new attribute.
 - enter a name for the attribute in the **Name** field.
 - check the **Primary UID** checkbox if the attribute is part of a primary key.
 - check the **Mandatory** checkbox if the attribute cannot be null.
Note: An * appears to the left of the attribute when this box is checked.

- ❑ The attributes that have been defined for an entity are displayed inside the entity box as shown on the slide [Add Entity Attributes \(3\)](#).

Add Entity Attributes (2)

The screenshot shows the 'Entity Properties - Employee' dialog box. The left sidebar contains a list of properties: General, Attributes (selected), Unique Identifiers, Relationships, Subtypes, Volume Properties, Engineer To, Comments, Comments in RDBMS, Overlapping Attributes, Notes, Impact Analysis, Measurements, Change Requests, Responsible Parties, Documents, Dynamic Properties, User Defined Properties, Classification Types, and Summary.

The main area is titled 'Attributes' and has three tabs: Details, Overview, and UDP. The 'Details' tab is active, showing a table of attributes and a form for editing the selected attribute.

	Name	Data type
1	empNo	Unknown
2	name	Unknown
3	address	Unknown
4	salary	Unknown
5	hkid	Unknown

The 'Attribute Properties' section for the selected attribute 'empNo' includes:

- Name: empNo
- Data Type: ☒ Domain, ☐ Logical, ☐ Distinct, ☐ Structured, ☐ Collection
- Source Type: Unknown (dropdown), Preferred ☐
- ☒ Primary UID, ☐ Relation UID, ☒ Mandatory, ☐ Deprecated

At the bottom, there are tabs for 'Comments in RDBMS', 'Comments', and 'Notes'. The 'Comments' tab is active, showing a text area for comments.

Buttons at the bottom: OK, Apply, Naming Rules, Cancel, Help.

Add Entity Attributes (3)

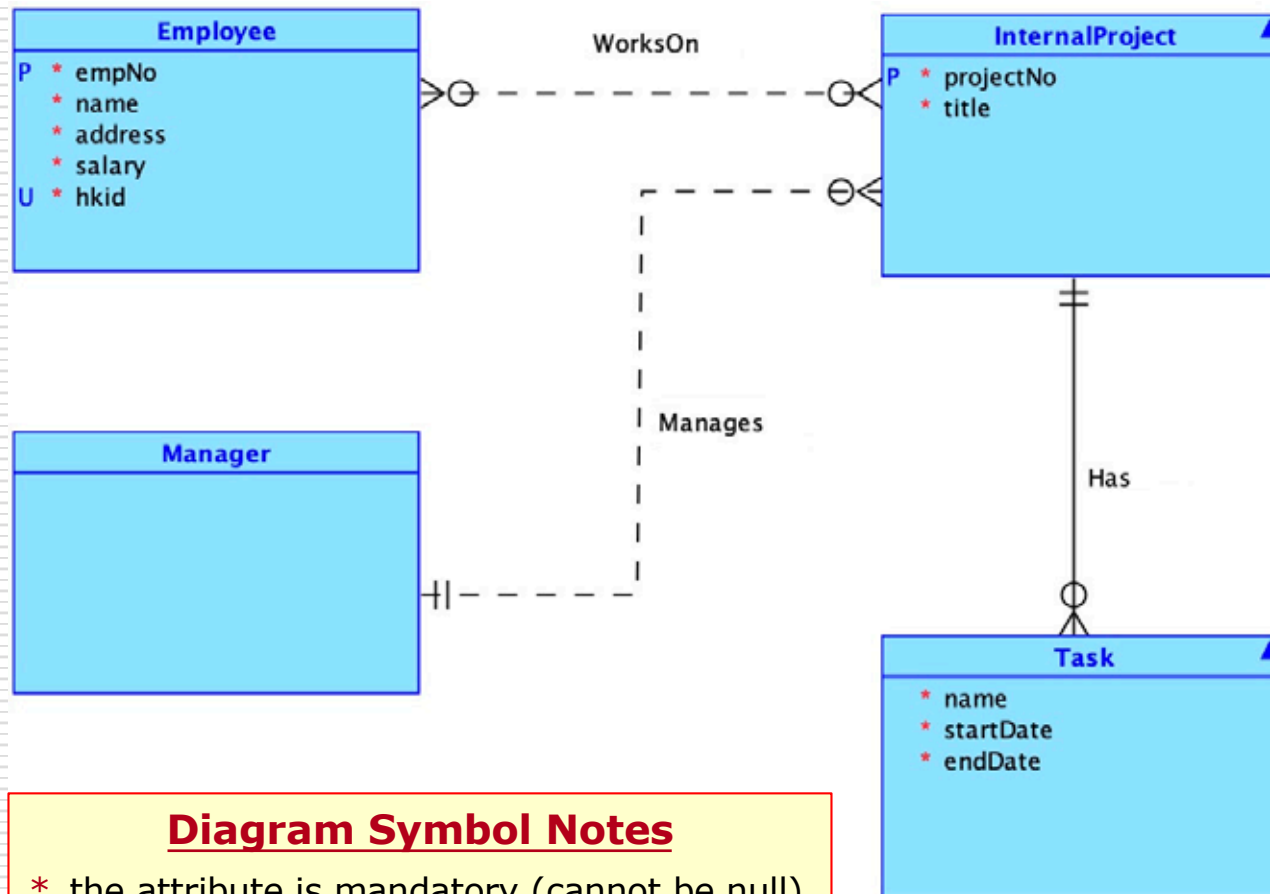
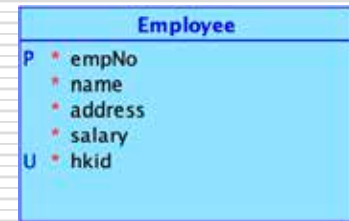


Diagram Symbol Notes

- * the attribute is mandatory (cannot be null)
- P the attribute is part of the primary key
- U the attribute's values are unique

Specify Unique Attributes

- ❑ To specify that an attribute's values are unique, but not a primary key, in the entity's **Entity Properties** dialog:
 - select the **Unique Identifiers** tab.
 - select **+** to specify a new unique attribute.
 - double-click the newly added attribute.
 - in the **Key Properties** dialog
 - ❑ select the **General** tab and edit the **Name** field to that of the unique attribute.
 - ❑ select the **Attributes and Relations** tab and select the unique attribute in the left pane and move it to the right pane.
 - select the **OK** button in the **Key Properties** dialog.
 - select the **OK** button in the **Properties** dialog.
- ❑ The attribute is displayed with a preceding **U** symbol as shown for the hkid attribute in the figure above and on the previous slide.



Add Relationship Attributes ⁽¹⁾

- ❑ Open the **Relation Properties** dialog for a relationship (see the slide [Create A Relationship Type \(2\)](#)) either by double clicking the relationship or right-clicking it and selecting **Properties** from the popup menu.
- ❑ In the **Relation Properties** dialog:
 - select the **Attributes** tab in the left column.
 - click **+** to add a new attribute.
 - enter a name for the attribute in the **Name** field.
 - select the **OK** button.
 - resize the attribute box as desired.
- ❑ The next slide shows an attribute defined for the **WorksOn** relationship.

Add Relationship Attributes (2)

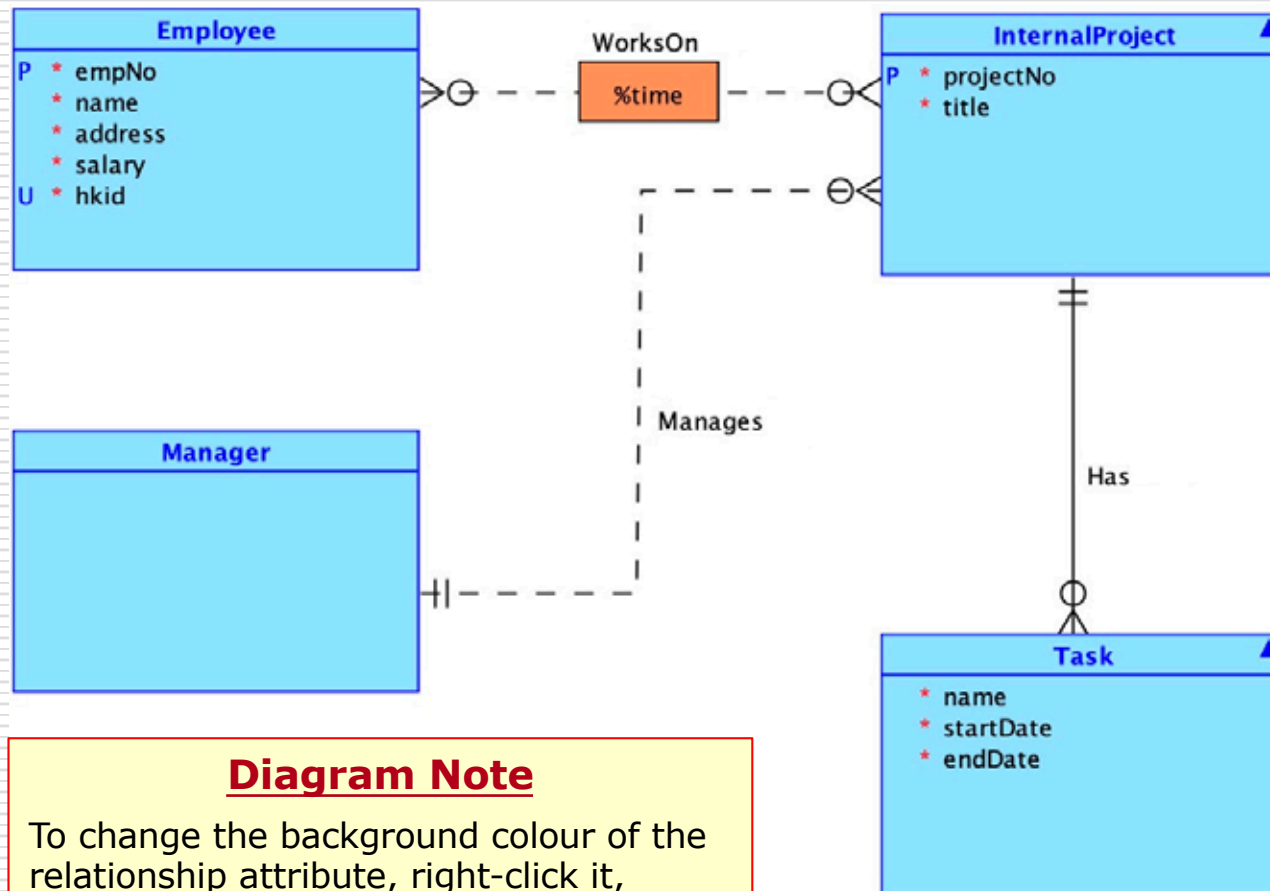


Diagram Note

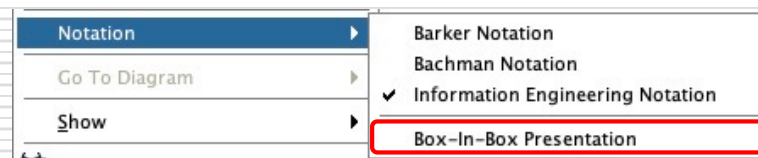
To change the background colour of the relationship attribute, right-click it, select **Format** from the popup menu and

- unselect **Use Default Format**.
- set **Background Color** to the desired colour.

Create A Generalization Relationship ⁽¹⁾

- ❑ Open the **Entity Properties** dialog for the subtype entity (e.g., **Manager**) and, in the **General** tab:
 - select the supertype in the **Super Type** dropdown list (e.g., **Employee** as shown on the next slide).
 - select the **OK** button.

Note: If **Box-In-Box Presentation** has not been unchecked in the **Notation** setting (see the slide [Oracle Data Modeler Design Settings](#)), then the supertype/subtype relationship will display as shown in the figure on the right. To display it as shown on the slide [Create A Generalization Relationship \(3\)](#), uncheck **Box-In-Box Presentation** in the **Notation** setting as shown in the figure below.



- ❑ The slide [Create A Generalization Relationship \(3\)](#) shows a generalization relationship example.

Create A Generalization Relationship (2)

The screenshot shows a software interface for defining entity properties. The window is titled "Entity Properties - Manager". On the left is a sidebar with a search icon and a list of categories: General, Attributes, Unique Identifiers, Relationships, Subtypes, Volume Properties, Engineer To, Comments, Comments in RDBMS, Overlapping Attributes, Notes, Impact Analysis, Measurements, Change Requests, Responsible Parties, Documents, Dynamic Properties, User Defined Properties, Classification Types, and Summary. The "General" category is selected. The main area is titled "General" and contains the following fields and controls:

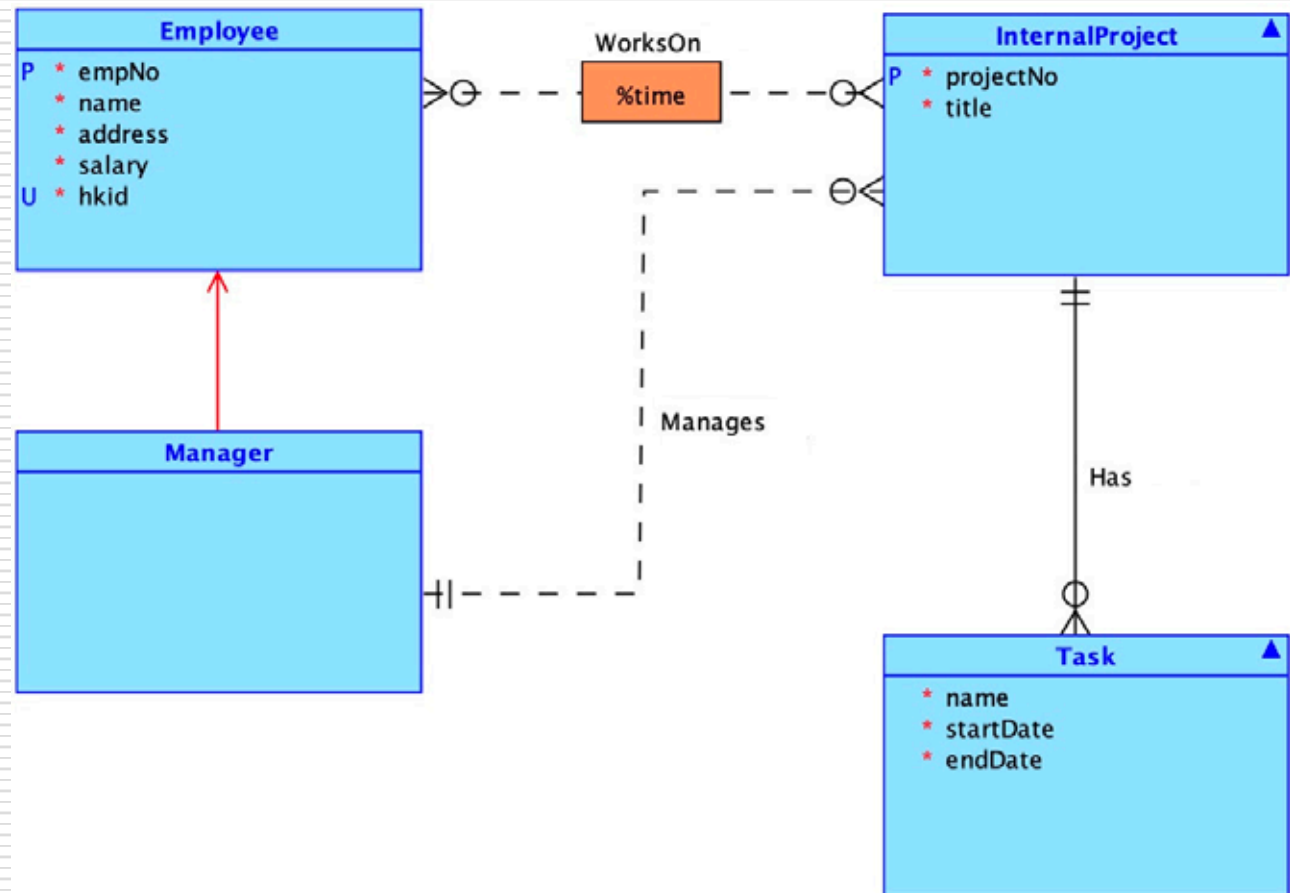
- Name: Manager
- Short Name: (empty)
- Synonyms: (empty)
- Synonym to display: (empty)
- Preferred Abbreviation: (empty)
- Long Name: Manager
- Based on Structured Type: (empty dropdown)
- Super Type: Employee (dropdown menu) with a "Select" button
- Source: (empty)
- Allow Type Substitution: ☒
- Create Surrogate Key: ☐
- Deprecated: ☐

At the bottom of the window are five buttons: OK, Apply, Naming Rules, Cancel, and Help.

Create A Generalization Relationship (3)

Reposition a generalization relationship by selecting it and dragging its endpoints to the desired positions.

Add elbows to a generalization relationship by right-clicking on it, selecting **Add Elbow** from the popup menu and dragging the elbow to the desired position.

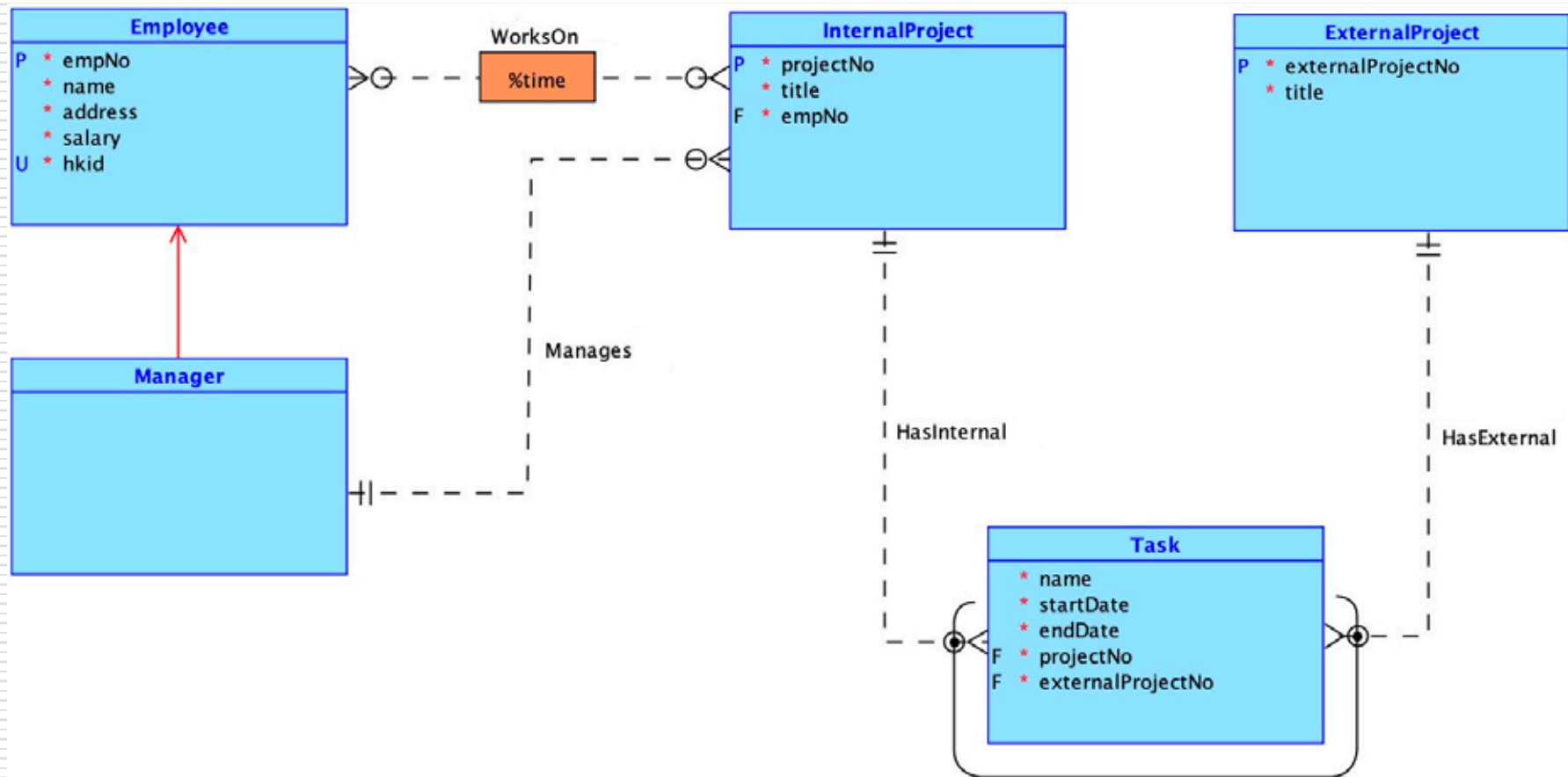


Create An XOR Constraint (Arc Relationship) ⁽¹⁾

- To create an XOR constraint (Arc relationship):
 - select the entity type that should participate in only one of the relationship types (e.g., Task).
 - select all relationships to be included (hold down the Shift key and select each relationship).
 - select the **New Arc** button in the toolbar.

- An example of an XOR constraint (Arc relationship) is shown on the next slide.

Create An XOR Constraint (Arc Relationship) (2)

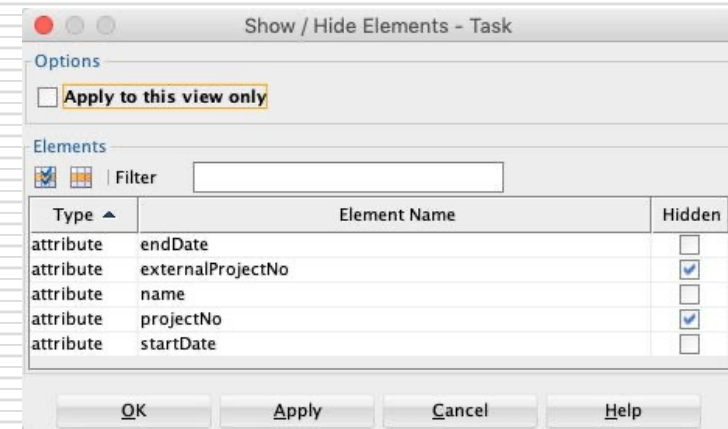


Showing/Hiding Design Elements (1)

- ❑ To hide unwanted elements, such as foreign keys, within an entity:
 - right-click on the entity and select **Show / Hide Elements** from the popup menu.
 - select the elements to hide.
 - select the **OK** button.

Note: When entity elements are hidden, a triangle ▲ appears in the upper right corner of the entity as shown on the next slide for the entities **Task** and **InternalProject**.

- ❑ The next slide shows an example of hiding some entity elements (compare with the previous slide).



Showing/Hiding Design Elements (2)

